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MILITARY SPECIFICATION

CYLINDERS, STEEL, COMPRESSED GAS, NON-SHATTERABLE, WELDED,

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the requirements for 1800 psi (12,400 kPa) and 2100 psi (14,500 kPa) (rated pressure) non-shatterable steel cylinders of welded construction.

* 1.2 Classification. Cylinders shall be of the following types and classes. The specific type and class shall be as specified by the applicable part number (see 6.2.d).

1.2.1 Types.

Type A - Internal thread
Type B - External thread

1.2.2 Classes.

Class 1 - 1800 PSI (12,400 kPa) rated pressure
Class 2 - 2100 PSI (14,500 kPa) rated pressure

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2b).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Code SR3, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8120
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SPECIFICATIONS

FEDERAL

BB-A-1034	-	Compressed Air, Breathing.
BB-N-411	-	Nitrogen, Technical.
TT-C-490	-	Cleaning Methods for Ferrous Surfaces and Pretreatment for Organic Coatings.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-wall.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-C-5501	-	Caps and Plugs, Protective, Dust and Moisture Seal.
MIL-O-27210	-	Oxygen, Aviator's Breathing, Liquid and Gas.
MIL-T-27730	-	Tape, Antiseize, Tetrafluoroethylene, with Dispenser.
MIL-M-43719	-	Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, General Specification for.
MIL-C-81302	-	Cleaning Compound, Solvent, Trichlorotrifluoroethane.

STANDARDS

MILITARY

MIL-STD-101	-	Color Code for Pipelines and Compressed Gas Cylinders.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-453	-	Inspection, Radiographic.
MIL-STD-810	-	Environmental Test Methods & Engineering Guidelines
MIL-STD-1264	-	Radiographic Inspection for Soundness of Welds in Steel by Comparison to Graded ASTM E 390 Reference Radiographs.

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MIL-STD-1595	-	Qualification of Aircraft, Missile and Aerospace Fusion Welders.
MIL-STD-2219	-	Fusion Welding for Aerospace Applications.
MS90389	-	Cylinders, Oxygen, Non-Shatterable, Welded, Steel.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

1684AS100	-	Cylinder, Oxygen Non-shatterable, Welded (1,800 psi).

(Copies of drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

PUBLICATIONS

Defense Standardization Manual

SD-6	-	Provisions Governing Qualification

(Copies of defense standardization manuals are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2b).

COMPRESSED GAS ASSOCIATION (CGA)

Pamphlet No. C-1	-	Methods for Hydrostatic Testing of Compressed Gas Cylinders.

(Application for copies of CGA pamphlets should be addressed to the Compressed Gas Association, Inc., Crystal Gateway 1, Suite 501, 1235 Jefferson Davis Highway, Arlington, VA 22202.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8-89 - Tension Testing of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations thatprepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Qualification. The cylinders furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3, 4.3.1, and 6.3). In addition, the retention of the qualification for the cylinders on the applicable qualified products lists shall be dependent on periodic verification of continued compliance with the requirements of this specification (see 4.3 and 4.3.2).

* 3.2 First Article. When specified (see 6.2h), the samples shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

* 3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practical weight, and suitable for the purpose intended. Cylinders may be made from pierced billets, cupped plates, or seamless tubing with either welded pressure heads or spun bottoms. The pressure heads used with tubes from a particular heat shall also be manufactured from a single heat in order to simplify gunfire testing.

3.3.1 Steel. Cylinders shall be made of austenitic type corrosion resistant steel in accordance with any one of the chemical analyses shown in Table I for essential elements.

* 3.4 Design and construction. The design and construction of the cylinders shall be in accordance with the applicable part number of Drawing 1684AS100 or MS90389 as specified in the contract or order (see 6.2d) and shall be of welded construction. The cylinders shall have been subjected to the initial permanent volumetric expansion test within 18 months prior to delivery.

* 3.4.1 Wire wrapping. Unless otherwise specified in the contract or applicable drawings, wire wrapping shall not be used in the construction of cylinders furnished under this specification.

* 3.4.2 Welding. Welding shall be performed in accordance with MIL-STD-2219 by a welder qualified in accordance with MIL-STD-1595. Welded areas shall be subjected to radiographic inspection.

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3.4.3 Neck. Shrink rings or supporting collars shall not be used for strengthening the neck of the cylinder.

3.4.4 Closure. Unless otherwise specified herein, the threaded port of each cylinder shall be closed by a metal, removable, threaded plug or cap conforming to MIL-C-5501 or a commercially equal metal plug or cap, to protect the threads and prevent entrance of foreign matter or moisture.

3.5 Performance.

3.5.1 Visual examination and dimensional check.

3.5.1.1 Visual examination. When visually examined as specified in 4.7.1.1, the cylinders shall conform to the requirements of this specification. Table V, Classification of Defects for Visual Examination, shall be used to classify and enumerate the defects.

3.5.1.2 Dimensional check. When dimensionally checked as specified in 4.7.1.2, the cylinders shall conform to the requirements of Drawing 1684AS100 or MS90389, as applicable. Any measurement deviating from the specified dimensions and tolerances shall be classified as a dimensional defect.

3.5.2 Cleanliness.

3.5.2.1 Non-volatile residue. When tested as specified in 4.7.2.1, the residue from the solvent used for the rinse shall be not greater than that of the clean solvent by more than 0.0015 grams per square foot of part surface area.

3.5.2.2 Particulate test. When tested as specified in 4.7.2.2, the solvent rinse shall not contain more than 5 particles between 100 and 175 micrometers per 100 ml of solvent; and no particles larger than 175 micrometers per 100 ml of solvent.

3.5.2.3 Halogen test. When tested as specified in 4.7.2.3, there shall be no evidence of cleaning compound residue.

3.5.3 Leakage. When tested as specified in 4.7.3, the cylinder shall not show any evidence of gas bubbles after 2 minutes of submergence.

3.5.4 Permanent volumetric expansion. When tested as specified in 4.7.4, the cylinder shall not have a permanent volumetric expansion greater than 10 percent of the total volumetric expansion.

* 3.5.5 Burst pressure. When tested as specified in 4.7.5, class 1 cylinders shall not burst at a pressure less than 4,500 psi (31,030 kPa) and class 2 cylinders shall not burst at a pressure less than 5,250 psi (36,200 kPa). Failure of cylinders shall occur in a ductile manner.

3.5.6 Endurance. When tested as specified in 4.7.6, the cylinder shall not show any evidence of material failure.

3.5.7 Flattening. When tested as specified in 4.7.7, the cylinder shall show no evidence of cracking when examined visually without any magnification.

3.5.8 Macrostructure defects. When tested as specified in 4.7.8, the cylinder shall show no evidence of abnormal segregation, pipes, cracks, seams or abnormal change in structure from the surface to the center.

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3.5.9 Radiographic inspection. When tested as specified in 4.7.9, welded sections of the cylinder shall be in accordance with MIL-STD-1264.

3.5.10 Fragmentation resistance.

3.5.10.1 Procedure I. When tested as specified in 4.7.10.1, the cylinder shall remain in one piece.

3.5.10.2 Procedure II. When tested as specified in 4.7.10.2, the cylinder shall be allowed to separate into two pieces. The projected area of any piece shall not be less than 2.0 square inches. The cylinder shall exhibit no evidence of shattering with small irregular pieces or splinters.

* 3.5.11 Vibration (spiral cylinder). When tested as specified in 4.7.11, the cylinder assembly (brackets and cylinder) shall remain in one piece, and there shall be no evidence of constructural or material failure.

3.5.12 Physical properties. When tested as specified in 4.7.12, the samples shall not have a tensile strength greater than 200,000 psi (1.38×10^6 kPa) and elongation shall be not less than 6 percent.

* 3.6 Weight. The weight of the cylinder shall be not greater than the values specified on the applicable drawings.

3.7 Color. Unless otherwise specified on the applicable drawings or the acquisition document (see 6.2m), the color of the cylinders shall be in accordance with MIL-STD-101.

3.8 Finish. The external and internal surfaces shall be cleaned in accordance with Method I or IV of TT-C-490. The internal surface of cylinders fabricated of material other than corrosion-resistant steel shall be treated in accordance with Type I of TT-C-490. The external surface of all cylinders shall be painted with a protective finish that will not crack, chip, or scale during normal service life or under extreme environmental conditions. The color of the protective finish shall be as specified in 3.7.

3.9 Identification marking. The cylinders shall be marked for identification in accordance with MIL-STD-130.

3.9.1 Standard Markings. The cylinders shall be permanently and legibly marked using a pressure sensitive decal located as shown on Drawing 1684AS100 and MS90389, as applicable. The adhesive on the decal shall conform to MIL-M-43719, Type I, Class I. The markings shall be not greater than 1/8 inch (3.18 mm) high and shall consist of the following (listed sequence mandatory):

a. Manufacturer's Name and CAGE (Commercial and Government Entity) code.

b. Government Part Number.

c. 1800 psi (12,400 kPa) or 2100 psi (14,500 kPa), as applicable.

d. Serial Number.

e. Date of permanent volumetric expansion test (month and year) with identifying symbol between the month and year.

g. A blank area for the addition of subsequent PVE test dates.

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3.9.2 Additional markings. The following additional markings shall be applied by stencilling or clear pressure sensitive decal in two places diametrically opposite of each other (see applicable drawings for location):

Aviator's Breathing Oxygen

1800 PSI (12,400 kPa) or 2100 PSI (14,500 kPa)(as applicable)

The size of the markings and the space between each row of markings shall be not less than 0.12 times the outer diameter of the cylinder. The adhesive on the decal shall conform to MIL-M-43719, Type I, Class I.

3.10 Workmanship. The cylinders shall be uniform in quality and shall be free from irregularities or defects which could adversely affect safety, performance, reliability or durability. The cylinders shall be free of oil, grease, fuel, water, dust, dirt or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. First article inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in Table II.

4.3.1 Qualification samples. Qualification samples shall consist of the following:

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a. Nine cylinders of the type (Part Number) for which qualification testing has been authorized.

b. Four material samples in accordance with ASTM E8, cut from 4 completed cylinders.

c. Two sets of manufacturer's drawings in accordance with SD-6.

d. When applicable, certification from the original manufacturer that the distributor requesting qualification status is authorized to distribute the item, or to rebrand and distribute the item, as applicable.

Samples shall be forwarded to the test facility set forth in the letter of authorization to submit samples (see 6.4). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-C-29576A and number under authorization (reference authorizing letter and number (see 6.4).

4.3.2 Retention of Qualified Products Listing (QPL). The retention of qualification listings shall consist of verification every two years to determine compliance of the listed item with the requirements of this specification. Verification shall be by manufacturer's certification unless otherwise specified by the activity responsible for the QPL.

4.4 First article inspection. First article inspection shall consist of the examinations and tests specified in Table III, Listed Sequence Mandatory. If there is no Qualified Products Listing for the part number being procured, first article testing shall consist of the tests specified in Table II.

4.4.1 First article samples. Unless otherwise specified herein, as soon as practicable after award of a contract or order, the manufacturer shall submit four cylinders of each part number specified in the contract or order. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these cylinders from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2h). Approval of the first article samples or the waiving of the first articles inspection does not exclude the requirements of submitting to the quality conformance inspection. The first article samples shall be furnished to the Government as specified in the contract (see 6.2i).

4.4.1.1 First article information. Upon completion of the first article inspection program, the Government activity responsible for conducting the inspection program (see 6.2i) shall report to the contracting officer the results of the inspection program, with appropriate recommendations.

4.4.1.2 First article sample disposition. The samples will be destroyed during the inspection program and will not be returned to the contractor.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Table IV (Listed Sequence Mandatory). The sampling and inspection levels and acceptance criteria shall conform to MIL-STD-105.

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4.5.1 Sampling.

4.5.1.1 Inspection lot.

4.5.1.1.1 Cylinder. An inspection lot size shall be expressed in units of one cylinder of the same size, made essentially under the same conditions and from the same materials and components. The sample unit shall be one cylinder.

4.5.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing cylinders fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing cylinders fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of cylinders. The sample size, acceptance criteria, tests, and examinations required for the cylinders shall be as specified in Table IV.

4.6 Test conditions.

4.6.1 Gas. Unless otherwise specified herein, the gas used in testing the cylinders shall be either oxygen conforming to MIL-O-27210, Type I, or nitrogen conforming to BB-N-411, Type I, Class I, Grade B, or air conforming to BB-A-1034.

4.6.2 Temperature and pressure. Unless otherwise specified herein, tests shall be conducted at local ambient temperature and barometric pressure. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests. The temperature and barometric pressure shall be recorded at the time of inspection and, when required, the tests results shall be corrected to normal temperature and pressure (NTP) conditions. NTP conditions are 70°F (21.1°C) and 29.92 inches of mercury (101.3 kPa).

4.6.3 Antiseize tape. To prevent damage to the cylinder threads or valves, male pipe threads of fittings installed in cylinders during tests shall be properly coated with antiseize tape conforming to MIL-T-27730 before being installed in the cylinder. The tape shall be applied according to instructions specified in MIL-T-27730. Tape shall be prevented from entering the inside of the fittings.

4.7 Inspection methods.

4.7.1 Visual examination and Dimensional Check.

4.7.1.1 Visual examination. Every cylinder shall be visually examined for conformance to the requirements of 3.5.1.1. Defects shall be as specified in Table I.

4.7.1.2 Dimensional check. The cylinders shall be dimensionally checked for conformance to the requirements of 3.5.1.2.

4.7.1.3. Packaging. Each of the fully prepared shipping containers containing cylinders selected as a sample unit from the lot shall be examined to determine that the packaging, packing, and marking conform to Section 5 of this specification. The classification of defects, Table VI, shall be used to classify the defects found.

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4.7.2 Cleanliness (oxygen cylinders). The cleanliness test shall consist of the non-volatile residue test (see 4.7.2.1), the particulate test (see 4.7.2.2), and the halogen test (see 4.7.2.3).

4.7.2.1 Non-volatile residue test. To perform this test, place a clean plug in the outlet of the cylinder and clean the surface around the plug and cylinder outlet with cleaning compound solvent conforming to MIL-C-81302, Type I. Wipe dry with a clean, lint free cloth. Calculate the internal surface area of the cylinder. Using the internal surface area of the cylinder and a ratio of 200 ML per sq. ft. (2.15 L/M2), calculate the required amount of solvent (MIL-C-81302, Type I) to be used. In no case shall less than 150 ML of solvent be used. Remove the plug from the cylinder and pour in the solvent. Install the plug in the cylinder to seal the outlet. Agitate the cylinder for not less than ten minutes, in a manner to wash the entire internal surface. Immediately after completing the internal wash cycle, remove the plug and pour the solvent from the cylinder into a beaker. One-half of the solvent used for the rinse shall be used for determining weight of residue and the other half of the solvent shall be used for determining particle content (see 4.7.2.2). The weight of the residue shall be determined by separately evaporating to dryness one-half of the solvent used for rinse and an equal amount of clean solvent. When tested in this manner, the cylinders shall conform to the requirements of 3.5.2.1. Upon successful completion of this test, the remaining one-half of the solvent shall be subjected to the test specified in 4.7.2.2.

4.7.2.2 Particulate test. Upon successful completion of the test specified in 4.7.2.1, examine the remaining one-half of the solvent used for the rinse for particles. To determine the particle content, filter the solvent through a 1.2 micro milipore, or equivalent, membrane filter and inspect the filter surface for conformance to the requirements of 3.5.2.2. Upon successful completion of this test, the cylinders shall be subjected to the test specified in 4.7.2.3.

4.7.2.3 Halogen test. Upon successful completion of the test specified in 4.7.2.2, the cylinders shall be tested to detect any solvent residue. A Halogen or Halide detector, having a minimum sensitivity of not less than 3×10^{-4} standard cc's per second, shall be used for this test. The cylinders shall conform to the requirements of 3.5.2.3.

4.7.3 Leakage. The cylinder shall be charged to rated service pressure, 1800 +50,-0 psig (12,400 +345,-0 kPa) or 2100 +50,-0 psig (14,500 +345,-0 kPa). With this pressure maintained, the cylinder shall be submerged in water for 2 minutes, minimum. All parts of the cylinder shall be covered with not less than 1 inch (25.4 mm) of water. The cylinder shall pass the requirements specified in 3.5.3.

4.7.4 Permanent volumetric expansion. Permanent volumetric expansion of the cylinder shall be determined as specified in the hydrostatic test of Pamphlet C-1 of the Compressed Gas Association. The cylinder shall pass the requirements specified in 3.5.4. The cylinder shall have passed this test within 18 months prior to delivery.

* 4.7.5 Burst pressure.

4.7.5.1 Class 1 cylinders. The cylinder shall be hydrostatically pressurized at a rate not greater than 25,000 psig (172,650 kPa) per minute to a maximum pressure of 4350 psig (29,989 kPa) and the pressure maintained for three minutes. The pressure shall then be gradually increased to destruction and the burst pressure recorded. The cylinder shall conform to the requirements of 3.5.5.1.

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4.7.5.2 Class 2 cylinders. The cylinder shall be hydrostatically pressurized at a rate not greater than 25,000 psig (172,650 kPa) per minute to a maximum pressure of 5000 psig (34,530 kPa) and the pressure maintained for three minutes. The pressure shall then be gradually increased to destruction and the burst pressure recorded. The cylinder shall conform to the requirements of 3.5.5.2.

4.7.6. Endurance test. The cylinder shall be hydrostatically charged to 1800 +50,-0 psig (12,400 +345,-0 kPa) or 2100 +50,-0 psig (14,500 +345,-0 kPa), as applicable, and the pressure released to obtain an internal pressure of 50 ± 50 psig (345 ± 345 kPa). This shall constitute one cycle. The charging and discharging shall be repeated for 20,000 cycles. The cylinder shall pass the requirements specified in 3.5.6. The cylinder shall then be subjected to and pass the permanent volumetric expansion test.

4.7.7. Flattening. The cylinder shall be flattened to six times the wall thickness between wedge-shaped knife edges with a 60-degree (1.05 rad) included angle and rounded to a 1/2 inch (13mm) radius. The knife edge shall be applied perpendicularly to the longitudinal axis of the cylinder at the center of the cylinder. For longitudinal welds, the bend knuckle shall occur on the weld. The cylinder shall pass the requirements specified in 3.5.7.

4.7.8 Macrostructure (deep acid etch). Longitudinal sections from the cylinder shall be etched in an aqueous solution containing 50 percent concentrated hydrochloric acid by volume and maintained at a temperature of 160° ± 10° F (71° ± 5.5° C) for sufficient time to show the presence of defects. The sections shall then be visually examined by a metallographist and shall pass the requirements specified in 3.5.8.

4.7.9 Radiographic inspection. The welded sections of cylinders, except brackets welded to the spiral cylinder, shall be inspected in accordance with MIL-STD-453. The welded sections shall pass the requirements specified in 3.5.9.

4.7.10 Fragmentation resistance. Oxygen, conforming to MIL-O-27210, Type I (see 4.6.1), shall be used in the performance of this test.

4.7.10.1 Procedure I. Cylinders greater than 2-1/2 inches (6.35 cm) in diameter shall be subjected to gunfire under the following conditions:

a. The cylinder shall be charged to 1800 ± 50 psig (12,400 ± 345 kPa) or 2100 ± 50 psig (14,500 ± 345 kPa), as applicable, at gun range ambient air temperature.

b. The cylinder may be supported but not constrained.

c. The ammunition shall be 50 caliber M-2, armor-piercing.

d. The range shall be 50 yards (45.7 m) maximum.

e. The various cylinders taken for test shall be tested, each progressively, in a different position. The point of impact shall be near the weld. The cylinder positions shall be as follows:

(1) With the longitudinal axis of the cylinder normal to the line of fire.

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(2) With the longitudinal axis of the cylinder 45 degrees (.785 rad) from normal toward the gun position.

(3) With the longitudinal axis of the cylinder parallel to the line of fire with inlet port away from the gun position.

f. All shots shall be tumbled. A suggested method for projectile tumbling is shown on Figure 1.

g. The tumbled projectile shall have a velocity of not less than 2600 ft/sec (792 m/s) at the point of impact with the cylinder.

h. To verify that the projectile was satisfactorily tumbled, the size entry hole made by the projectile shall be not less than one-half inch by one and one-half inch (13 mm by 39 mm).

i. Verify that the projectile trajectory and tumble are satisfactory. This may be determined by the location and visual appearance of a hole made by the passage of the projectile through vertically suspended sheets of paper at the target area.

The cylinder shall pass the requirements specified in 3.5.10.1.

4.7.10.2 Procedure II. Cylinders 2-1/2 inches (6.35 cm) or less in diameter shall be subjected to gunfire under the following conditions:

a. The cylinder shall be charged to 1800 ± 50 psig (12,400 ± 345 kPa) or 2100 ± 50 psig (14,500 ± 345 kPa) at gun range ambient air temperature.

b. The cylinder may be supported but not constrained.

c. The cylinder shall be positioned as indicated in 4.7.10.1e.

d. The ammunition shall be 30 caliber armor piercing, with velocity of 2800 ± 100 feet per second (854 ± 30 m/s).

e. The range shall be approximately 20 yards (18.3 m).

f. Shots shall not be tumbled.

The cylinders shall pass the requirements specified in 3.5.10.2.

4.7.11 Vibration (spiral cylinder). The vibration test shall be conducted only in the horizontal positions and only at room temperature. The cylinder shall be pressurized to 1800 ± 50 psig (12,400 ± 345 kPa) or 2100 ± 50 psig (14,500 ± 345 kPa), as applicable, mounted to a rigid fixture by the cylinder's mounting brackets and then subjected to the vibration test specified in MIL-STD-810, Method 514.3, Procedure 1, Table 514.3-III, Figures 514.3-26 and 514.3-27. The power spectral density (Wo) shall be 0.30 G^2/HZ. The test time shall be 3 hours. The cylinder brackets shall then be examined and pass the requirements specified in 3.5.11.

4.7.12 Physical properties. Physical property testing of material samples shall consist of tensile strength and elongation testing in accordance with ASTM E8 for conformance to the requirments of 3.5.12.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C, as specified in the contract (see 6.2k).

5.1.1 Level A. Each cylinder shall be preserved and packaged in accordance with Method IC of MIL-P-116, without the use of contact preservatives. Cleaning shall be accomplished without the use of petroleum derivative solvents. The cylinder port shall be sealed with a metal cap or plug conforming to MIL-C-5501 or commercially equal metal cap or plug. When installing aluminum caps or plugs, caution shall be exercised in the application and tensioning of same to preclude the shearing of particles from the softer metal fittings. After cleaning, package each cylinder in a close fitting box conforming to PPP-B-636. The container closure shall be accomplished in accordance with applicable procedures specified in the appendix of PPP-B-636.

5.1.2 Level C. The cylinder shall be individually packaged in a manner that will prevent physical damage or environmental deterioration during transit from the shipping point to the original consignee. As a minimum, the cylinder port shall be plugged or capped, as applicable, with metal fittings only, conforming to MIL-C-5501, or commercial equal. When installing aluminum caps or plugs, caution shall be exercised in the application and tensioning of same to preclude shearing particles off the softer metal fittings.

5.2 Packing. Packing shall be Level A, B or C, as specified by the contract (see 6.2k). Insofar as practicable, shipping containers shall effect a close fit, contain identical quantities and be of uniform dimensional configuration. For Levels A and B, the gross weight of boxes shall be not greater than 200 pounds (90.7 kg).

5.2.1 Level A. A cylinder packaged as specified in 5.1.1 shall be packed for shipment in a Fiberboard Corrugated, Triple Wall box conforming to PPP-B-640, Class 2 (Weather Resistant), Style E. The closure sealing and reinforcing (banding) of the container shall be accomplished in accordance with the procedures specified in the appendix of the container specification.

5.2.2 Level B. A cylinder packaged as specified in 5.1.1 or 5.1.2 shall be packed for shipment as specified in 5.2.1, except that the fiberboard shipping container conforming to PPP-B-640 shall be Class 1 (Nonweather Resistant), Style E.

5.2.3 Level C. A cylinder packaged as specified in 5.1.1 shall be packed for shipment in a manner that will afford protection against damage during transit from the shipping point to the original consignee. Containers used shall be in accordance with Uniform Freight Classification Rules, or regulations of other carriers applicable to the mode of transportation utilized.

5.3 Marking. Each unit and shipping container shall be marked in accordance with MIL-STD-129. In addition, the following or similar precautionary marking shall be required on each unit package and shipping container.

CAUTION

DO NOT ALLOW PETROLEUM CONTAMINANTS OF ANY KIND
TO BE USED/STORED ON OR ABOUT THESE CYLINDERS.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended Use. The cylinders covered by this specification are intended for use in storing compressed gas.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including amendments/revisions.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Applicable Qualified Products List (QPL), including date.

d. Government Part Number (see 3.4).

e. National Stock Number.

f. Applicable Government drawings, including revisions (see 3.4).

g. Quantity desired.

h. Whether first article inspection is waived (see 4.4.1).

i. The name and address of the first article inspection laboratory (see 4.4.1) and the name of the Government activity responsible for conducting the first article inspection program (see 4.4.1.1).

j. Cylinders subjected to destructive tests (see Table IV) should not be shipped as part of the acquisition quantity.

k. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

l. Marking requirements (see 5.3).

m. Color (see 3.7).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL-29576) whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the qualified products list is the Commander, Naval Air Systems Command, Department of the Navy, Washington, D.C. 20361-5110; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Warfare Center Aircraft Division Warminster, Code 6031, Warminster, PA, 18974-5000.

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6.3.1. Drawings. When requested, the manufacturer should submit engineering drawings and inspection reports in accordance with SD-6.

* 6.4 First article. When a first article inspection is required, the item(s) should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

* 6.5 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.1.1	DI-NDTI-80809	Test/Inspection Report	

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.6 Subject term (key word) listing.

Color
External threads
First article
Fragmentation resistance
Internal threads
Pressure, rated, 1800 psi
Pressure, rated, 2100 psi
Qualification
Radiographic inspection
Trichlorotrifluoroethane
Wire wrapping

6.7 International standardization. Certain provisions (3.9.2), of this specification are the subject of international standardization agreement (STANAG 3056). When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

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6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AV
Air Force - 99

Preparing activity:
Navy - AS

(Project 8120-1018)

Review activities:
Air Force - 11 and 71
DLA - GS
International interest (see 6.7)

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TABLE I. Chemical composition by percentage.

C (max)	Mn (max)	P (max)	S (max)	Si (max)	Mo (min)	Ni (min)	Cr (min)	Ti (min)
0.12	2.00	0.04	0.04	1.00	---	8.00	17.0	0.60
0.08	2.00	0.04	0.04	1.00	---	8.00	17.0	0.40
0.08	2.00	0.04	0.04	1.00	2.00	10.0	16.0	0.40

TABLE II. Qualification inspection.

Inspection	Requirement	Test Method
Visual examination	3.5.1.1	4.7.1.1
Dimensional check	3.5.1.2	4.7.1.2
Cleanliness (Oxygen cylinders)	3.5.2.1 3.5.2.2 3.5.2.3	4.7.2.1 4.7.2.2 4.7.2.3
Leakage	3.5.3	4.7.3
Permanent volumetric expansion	3.5.4	4.7.4
Burst pressure	3.5.5	4.7.5
Endurance	3.5.6	4.7.6
Flattening	3.5.7	4.7.7
Macrostructure defects	3.5.8	4.7.8
Radiographic inspection	3.5.9	4.7.9
Fragmentation resistance	3.5.10.1 or 3.5.10.2, as applicable	4.7.10.1 or 4.7.10.2, as applicable
Vibration (spiral cylinder)	3.5.11	4.7.11
Physical properties	3.5.12	4.7.12

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TABLE III. First article inspection.

Inspection	Requirement	Test Method
Visual examination	3.5.1.1	4.7.1.1
Dimensional check	3.5.1.2	4.7.1.2
Cleanliness (Oxygen cylinders)	3.5.2.1, 3.5.2.2, 3.5.2.3	4.7.2.1, 4.7.2.2, 4.7.2.3
Leakage	3.5.3	4.7.3
Permanent volumetric expansion	3.5.4	4.7.4
Burst pressure	3.5.5	4.7.5
Flattening	3.5.7	4.7.7
Radiographic inspection	3.5.9	4.7.9

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TABLE IV. Quality conformance inspection.

Inspection	Method	Sample Size	Acceptance criteria
Visual examination	4.7.1.1	a. Every cylinder for critical defects.	a. Reject all units with any critical defect.
		b. Inspection Level II for minor defects.	b. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensional check	4.7.1.2	Inspection Level S-2 1/	Acceptance number zero, rejection number 1.
Cleanliness	4.7.2.1, 4.7.2.2, 4.7.2.3	Inspection Level S-3 1/	Acceptance number zero, rejection number 1.
Leakage	4.7.3	Every cylinder	Reject all defective units.
Permanent volumetric expansion	4.7.4	Every cylinder	Reject all defective units.
Burst pressure 2/	4.7.5	Inspection Level S-2 1/	Acceptance number zero, rejection number 1.
Flattening 2/	4.7.7	Inspection Level S-1 1/	Acceptance number zero, rejection number 1.
Radiographic inspection	4.7.9	Every cylinder	Reject all defective units.
Physical properties 2/	4.7.12	Inspection Level S-1 1/	Acceptance number zero, rejection number 1.
Packaging	4.7.1.3	Inspection Level S-2	Total acceptable quality level of 4.0 percent defective.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.
2/ This is a destructive test (see 6.2j).

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TABLE V. Classification of defects for visual examination of the cylinder.

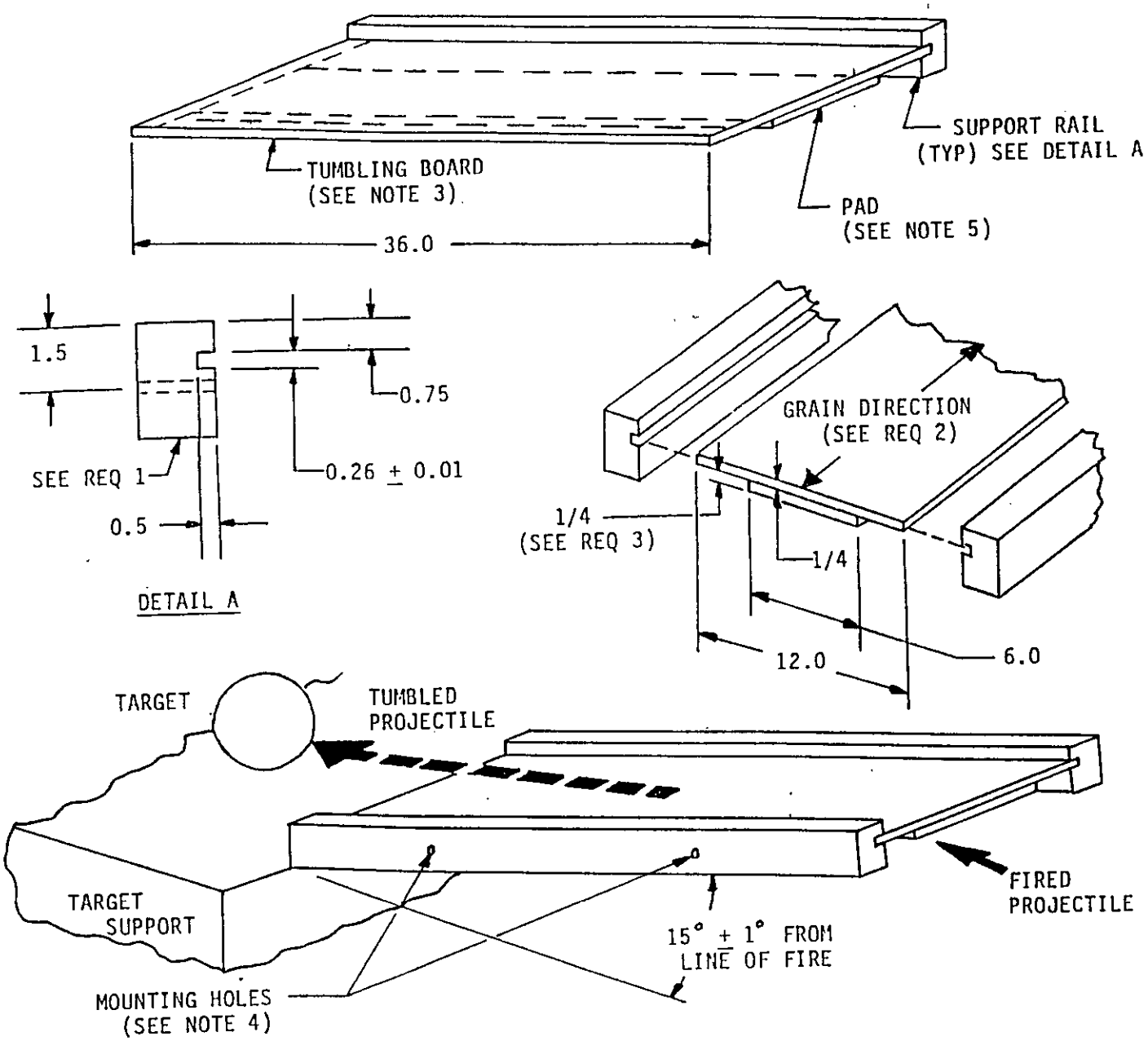
Defects	Critical	Minor
Incorrect outlet configuration	X	
Incorrect or defective threads	X	
Incorrect, incomplete, or faulty finish		X
Incorrect part number or cylinder code number	X	
Flaw in material	X	
Incorrect or incomplete marking		X
Incorrect color		X
Abnormally distorted or nonuniform weld	X	
Loose particles inside the cylinder	X	
Rust or scale	X	
Irregularities in the internal surface greater than 1/16 inch.	X	
Existence of any foreign material, solid or liquid.	X	

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TABLE VI. List of defects for packaging.

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials.	Any nonconforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content.	Number per container is more or less than required; gross or net weight exceeds the requirements.

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REQUIREMENTS:

1. LUMBER, 2x3 - FINISH SIZE
2. BOARD, PLYWOOD-GRADE A-A, EXTERIOR TYPE, 3 PLY
3. SHEET, RUBBER-STYRENE-BUTADIENE (SBR), 55 ± 5 DUROMETER (SHORE A)

NOTES:

1. DIMENSIONS IN INCHES
2. TOLERANCE: DECIMALS ± 0.1, FRACTIONS ± 1/64
3. TUMBLING BOARD MAY BE MOVED IN THE SUPPORT RAILS FOR REPEATED FIRINGS
4. MOUNTING HOLES METHOD USED TO SUPPORT TUMBLING BOARD IS OPTIONAL, HOWEVER, THE BOARD MUST BE RIGIDLY MOUNTED.
5. PAD SHALL BE CEMENTED TO ENTIRE LENGTH OF TUMBLING BOARD USING A COMMERCIAL CONTACT CEMENT

FIGURE 1. Projectile tumbling method

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-29576A	92/09/02

3. DOCUMENT TITLE
CYLINDERS, STEEL, COMPRESSED GAS, NON-SHATTERABLE, WELDED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	(908) 323-7488	624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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